

Deep in the Tails: Multi-Task Learning for Macroeconomic Tail Risk^{*}

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Abstract

We develop a deep multi-task neural network (DMQ) to forecast the conditional distribution of inflation, industrial production, and unemployment from 1998-2024. As benchmarks, we use quantile regressions from the latest macroeconomic risk literature and a large set of popular machine-learning methods. We find DMQ provides substantial gains in forecast accuracy for the conditional distribution as well as the conditional mean, especially during the GFC and Covid.

Keywords:

JEL Codes:

1 Introduction

In recent years policy makers have become increasingly concerned with uncertainty surrounding macroeconomic forecasts. For example, the Federal Open Market Committee (FOMC) regularly includes distributions around its predictions for key macroeconomic indicators

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within its outlook publications. Distributions around forecasts are important because point forecasts fail to paint an accurate picture of the level of risk in the economy. Policy makers are particularly concerned with tail risks and economic uncertainty clearly is factored in their decisions ([Andrade et al., 2012](#)).

2 Method

2.1 ...

3 Conclusion

References

ANDRADE, P., E. GHYSELS, AND J. IDIER (2012): “TAILS OF INFLATION FORECASTS AND TALES OF MONETARY POLICY,” *MONETARY POLICY*.

A Appendix